

Seals in Motion.

Flexible Sealing Solutions for the Mobility of the Future.



Thesis Progress Report

Niederwinkling, 29.02.24

Wen-Hao Wu

Agenda

- Thesis & Project Progress Report
- The Present & Future of AI in W&S

Thesis Report

Core Idea of my Thesis in one Sentence

- I want to create an AI that can detect defects „**better**“ than the current inspection machines
- What means „**better**“?

Thesis Report

What means „Better“?

- **Higher Accuracy**
 - No Escaped Defect (priority)
 - Less Pseudo Scrap
- **Faster**
 - Without sacrificing the cost of high-end hardware
- **Robust**
 - Provide consistent results

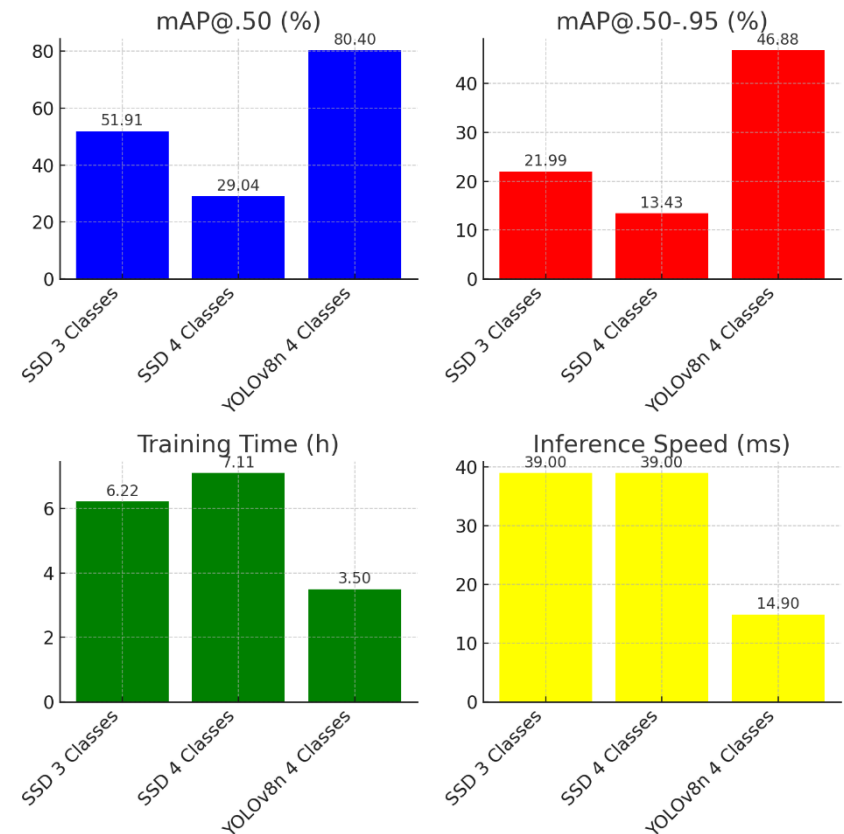
Does the part meet the quality requirements?

Real Condition	KO	True-Positive Correct Scrap	False-Negative Escaped Defect	KO
	OK	False-Positive Pseudo Scrap	True-Negative Good Parts	OK
		Test Result		

Thesis Report

My Approach to achieve „more Accurate and more Robust“

- Comparison of 2 different DL models
- YOLOv8 overall superior



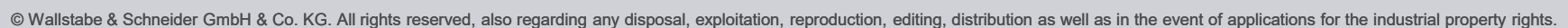
Thesis Report

My Approach to achieve „Faster“

- Detection using a GPU is always faster than a CPU
- GPUs are costly and CPUs are cheap
- The real question is: How to detect faster on a CPU?
- Solution: Convert YOLOv8 model to **ONNX** format (open-sourced by Microsoft)
- Inference speed is increased up to **3 times**

```
 ".PT GPU DETECTION": {  
   "Average Inference Speed": 11.31,  
   "Average Preprocess Speed": 7.25,  
   "Average Postprocess Speed": 5.69  
 },  
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   "Average Preprocess Speed": 6.55,  
   "Average Postprocess Speed": 3.37  
 },  
 ".ONNX CPU DETECTION": {  
   "Average Inference Speed": 54.56,  
   "Average Preprocess Speed": 6.65,  
   "Average Postprocess Speed": 5.81  
 }
```

- ~2200 pictures with careful markings



Thesis Report

What else was done?

Cut, Missing, Rough Surface, Excess Material

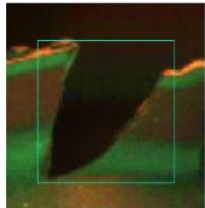


Figure 4.5: cut example 1

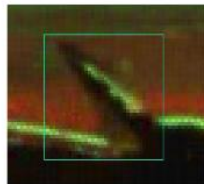


Figure 4.6: cut example 2

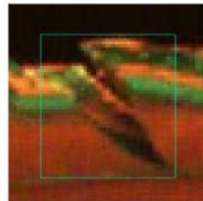


Figure 4.7: cut example 3

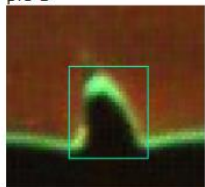


Figure 4.8: cut example 4

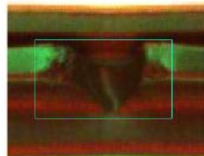


Figure 4.9: cut example 5



Figure 4.10: cut example 6

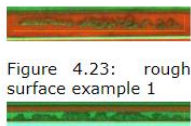


Figure 4.23: rough surface example 1

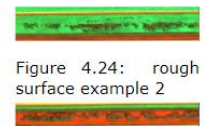


Figure 4.24: rough surface example 2

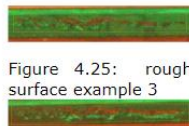


Figure 4.25: rough surface example 3

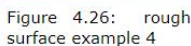


Figure 4.26: rough surface example 4

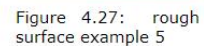


Figure 4.27: rough surface example 5

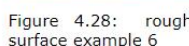


Figure 4.28: rough surface example 6

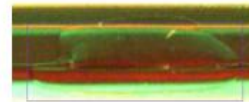


Figure 4.11: excess material example 1

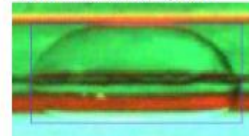


Figure 4.14: excess material example 4

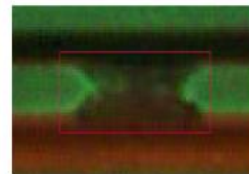


Figure 4.17: missing example 1



Figure 4.20: missing example 4



Figure 4.12: excess material example 2

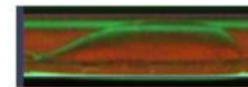


Figure 4.15: excess material example 5



Figure 4.18: missing example 2

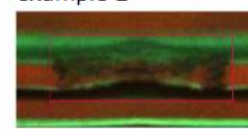


Figure 4.21: missing example 5

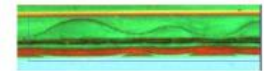


Figure 4.13: excess material example 3

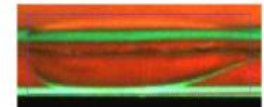


Figure 4.16: excess material example 6



Figure 4.19: missing example 3

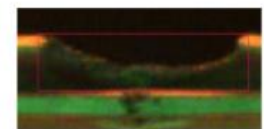
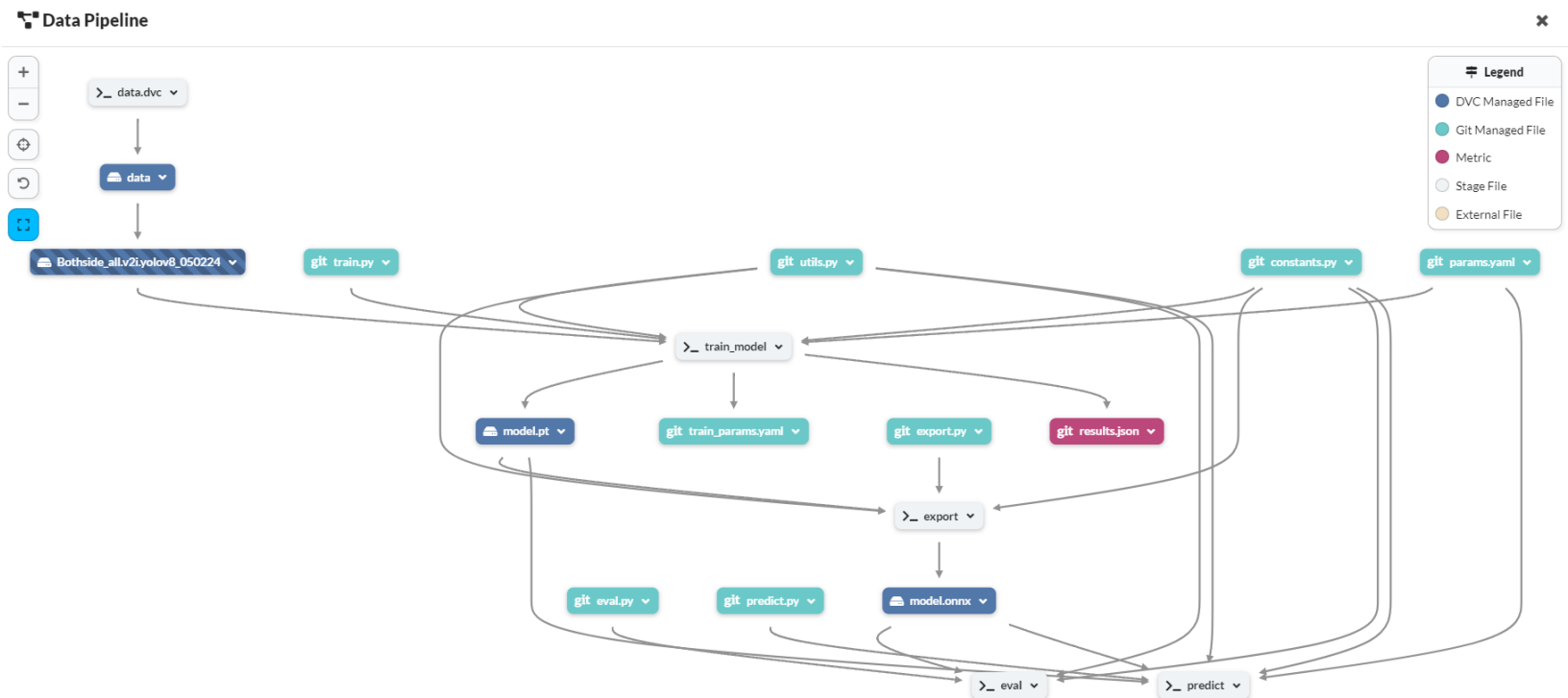


Figure 4.22: missing example 6

Thesis Report

What else was done?

- An operation **pipeline** that automates the ML Process (**Train, Export, Evaluation, Predict**)



Thesis Report

What else was done?

- **Store and index** training parameters and results in a cloud-based database

InternAI_WS / yolov8-defect-detection
connected to https://github.com/Gary0417/yolov8-defect-detection.git - 2 hours ago

Files Datasets 0 Experiments 14 Models Collaboration Annotations

Compare Reset filters Delete Archive Labels Columns Log experiment Go to MLflow UI

	Code	Name	Commit	Cre... ↓	Labels ▾	Sou...	device	workers	project	metrics/ma...	metrics/ma...	metrics/pre...
☐	✓		yolov8n-train-14	e909	9 days ago	mlflow	None	8	runs	0.4404203820...	0.7824696554...	0.8259134859...
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☐	✓		yolov8s-train-12	23a9	23 days ago	mlflow	None	8	runs	0.4867437734...	0.8078164721...	0.7994273479...
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☐	✓		yolov8m-train-6	8621	a month ago	mlflow	None	8	runs	0.2238457309...	0.4723849606...	0.5399652070...
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Thesis Report

What else was done?

- **Store and index** training parameters and results in a cloud-based database

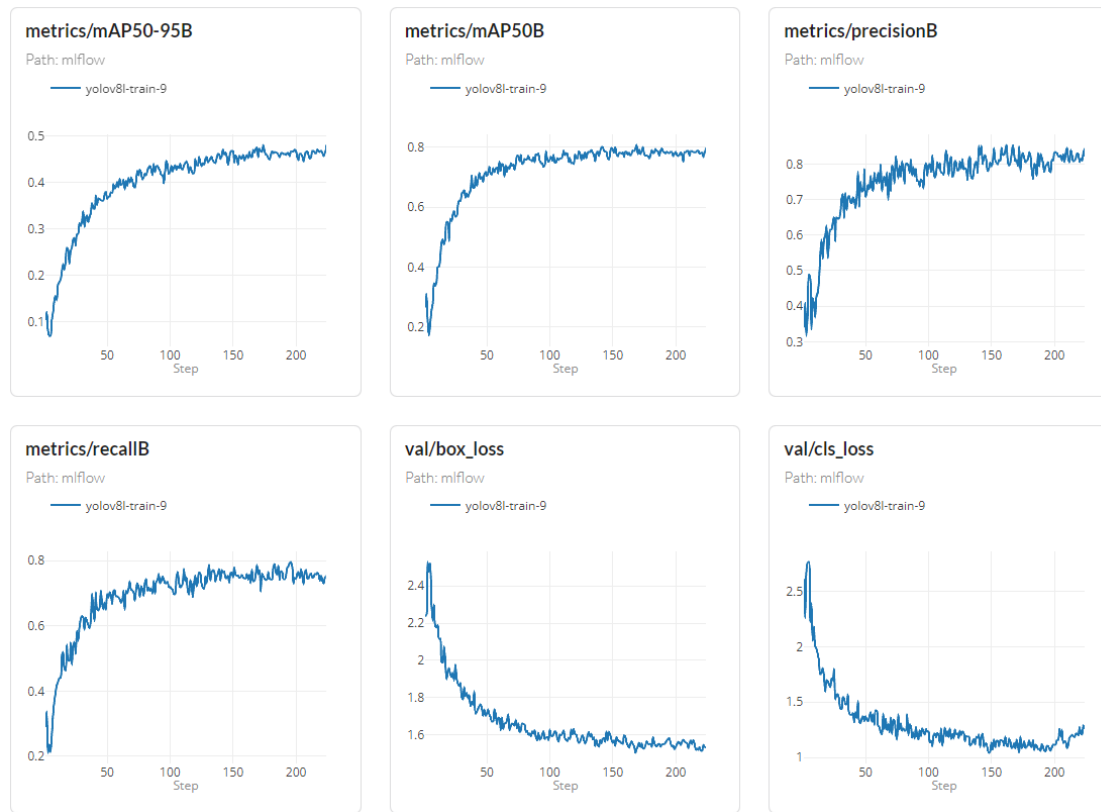
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Name	Value	
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amp	True	
augment	False	
batch	4	
box	7.5	
boxes	True	
cache	False	
cfg	None	
classes	None	
close_mosaic	10	
1 to 10 of 98		<< < Page 1 of 10 > >>

Metrics		Search Metrics...
Name	Value	
metrics/mAP50-95B	0.48	
metrics/mAP50B	0.799	
metrics/precisionB	0.821	
metrics/recallB	0.752	
val/box_loss	1.533	
val/cls_loss	1.257	
val/dfi_loss	1.132	
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Thesis Report

What else was done?

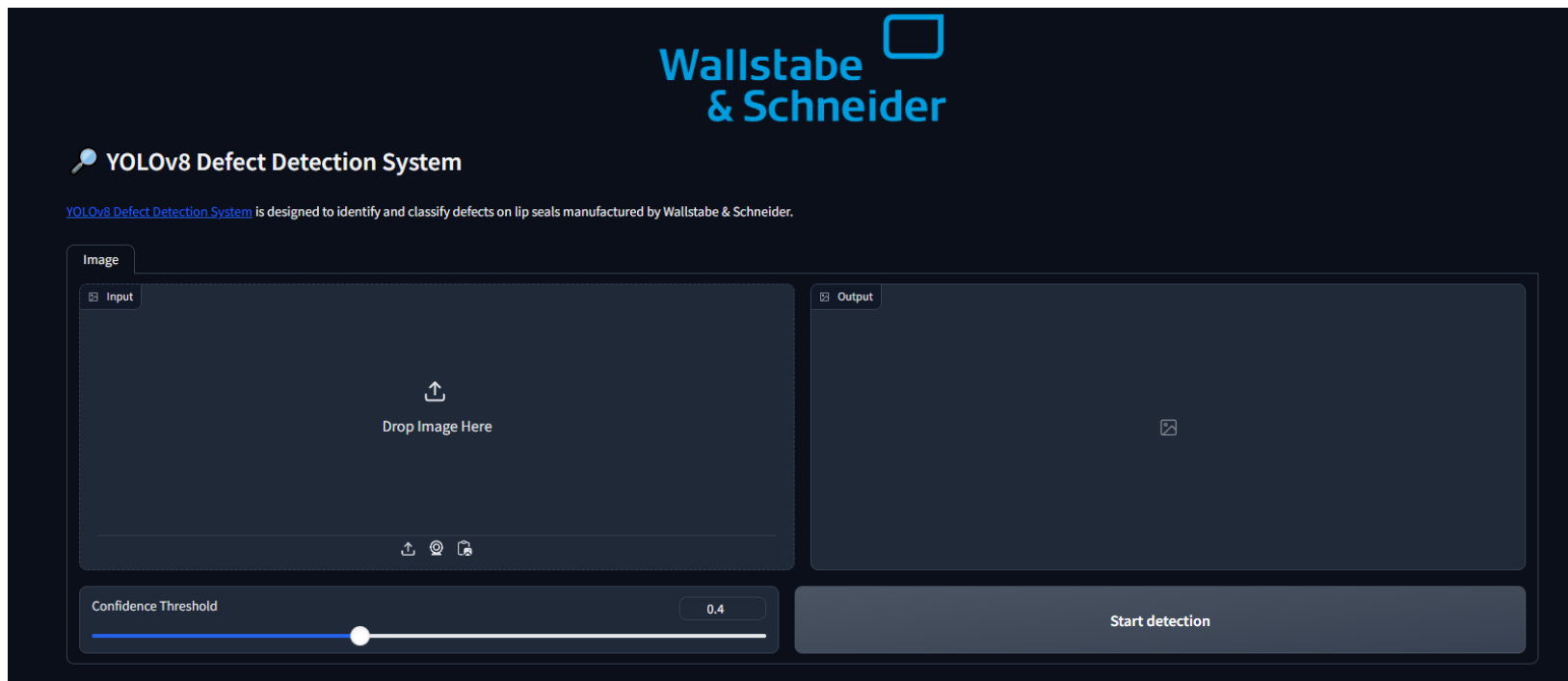
- **Store and index** training parameters and results in a cloud-based database



Thesis Report

What else was done?

- GUI for training and detection



Thesis Report

What else was done?

- GUI for training and detection

The screenshot shows a web-based GUI for training and detection. It features a dark theme with a sidebar on the left containing two tabs: 'Training' and 'Detection'. The 'Training' tab is currently selected. The main content area under 'Training' includes a 'Select Model' dropdown menu. Below this is a section for 'Data File Path (.yaml file)' which contains a large dark rectangle with an upload icon and the text 'Drop File Here - OR - Click to Upload'. Underneath are two sliders for 'Epochs Count' and 'Batch Size', both of which are set to the value 1. A 'Start Training' button is positioned below the sliders. At the bottom of the interface, there is a 'Training Status' text area and a 'Clear Temp Folder' button.

Thesis Report

What did I write in my Thesis?

- I documented the whole journey of my project of my approach to achieve „better“
 - Motivation and Objectives
 - Literature Reivew
 - Methodologies
 - Results Evaluation
 - Future Outlook

Thesis Report

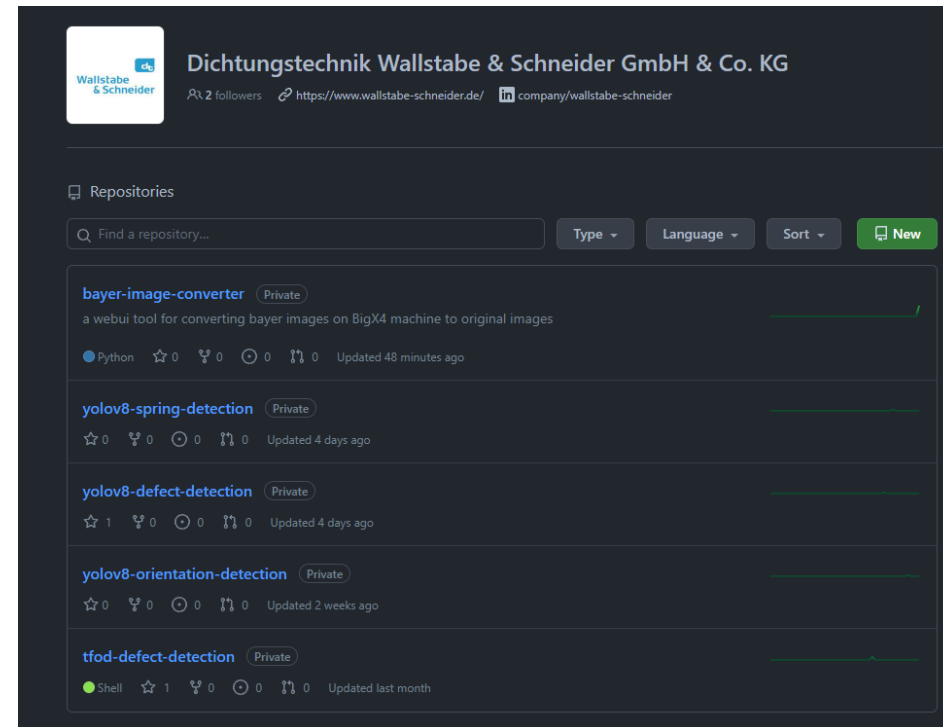
Future Outlook

- Add **more features** to the system:
 - Generating a report after detection
 - Further improve error handling
 - Further refine GUI
 - Further refine unit testing and validation testing
 - ..
- Further improve **dataset quality**
- Explore the opportunity to **scale the system** to detect on other articles
- Explore the methods to further **improve inference speed**
 - Deploying the system on a Linux environment
 - Convert the AI model to a different format
- Deploy the system onto **inspection platforms** (prototyping)
 - Cloud
 - Locally

The Present & Future of AI in W&S

Current AI Achievements

- A collection of developed software is stored on Github under the W&S page
 - Bayer-image-converter
 - YOLOv8-spring-detection
 - YOLOv8-defect-detection
 - YOLOv8-orientation-detection
 - Tfod-defect-detection (deprecated)
 - ...
- The source code and user guidelines are stored in each repository
- Documentation of UTP AI



The Present & Future of AI in W&S

Strategic Steps Towards the Future (my opinion)

- **AI team as a sub-department/ focus-team**
 - Cross-departmental collaboration
- **Leadership and organizational structure**
 - Senior AI-experienced leadership to guide the AI team
 - Establish best practices
 - Organize project management
 - Communicate as a bridge between different departments
 - Ensure strategic alignment of AI efforts with company goals
- **Work closer to the IT department**

The Present & Future of AI in W&S

Resource Allocation and Empowerment

■ Requirements:

- **Flexible** software installation and access to essential tools to address project needs and adapt to technological advancements

■ Secure Workstation Solutions:

- IT-Technik suggests building workstations outside the network to install software without risking network security.
- For needs like Gary's Bachelor Thesis, which requires network access and admin rights, the traditional approach is suboptimal.
- Proposed solutions:
 - **Virtual Machine**
 - **Dual Boot**
 - **VPN?**
- Additional suggestions from Heinz are to be considered.

The Present & Future of AI in W&S

Data Storage Challenges

■ Defect and Product Images:

- Utilize Roboflow for image annotation; it's efficient and free.
- Public vs. Private:
 - Continue with a public, free plan.
 - Opt for privacy at ~€250/month.

■ Experiment Data (AI Training):

- Stored in Dagshub (S3 bucket), outside company storage. Seeking opinions on this approach.

■ Output Data from Rubber Spring Project:

- Anticipated storage issue due to accumulating detection images.
- Potential solutions: External or cloud storage to manage the growing library.

Thank you!